



RIVER VALLEY HIGH SCHOOL

JC 2

PRELIMINARY EXAMINATION

CANDIDATE
NAME

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CENTRE
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CLASS

20J

INDEX
NUMBER

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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

16 September 2021

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your Centre number, index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together. The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	
2	
3	
Section B	
Total	

This document consists of **23** printed pages and **1** blank page.

Section A

Answer **all** the questions in this section.

- 1 (a) Discuss if cancer is a genetic disease. [3]

The transition of normal cells into the cancerous state is termed as cellular transformation and is accompanied by alteration in cell morphology as well as function. Fig 1.1 shows a group of cells, including cancer cells which has undergone cellular transformation.

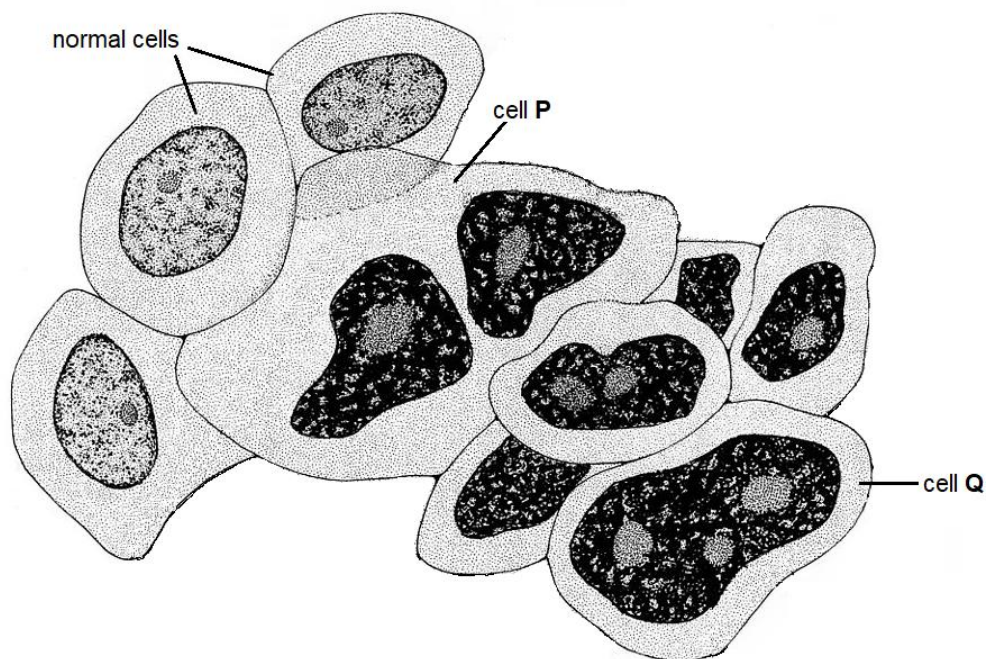


Fig. 1.1

- (b) Apart from cell shape and size, identify another structural abnormality visible in cell Q, and explain how this abnormality may contribute towards cancerous growth. [3]

- (c) Cells with morphology similar to cell **P** are common in cancer. These cells arose as a result of the failure of the mitotic cell cycle.

State the stage in which the mitotic cell cycle failed and explain how this led to the production of cell **P**.

[4]

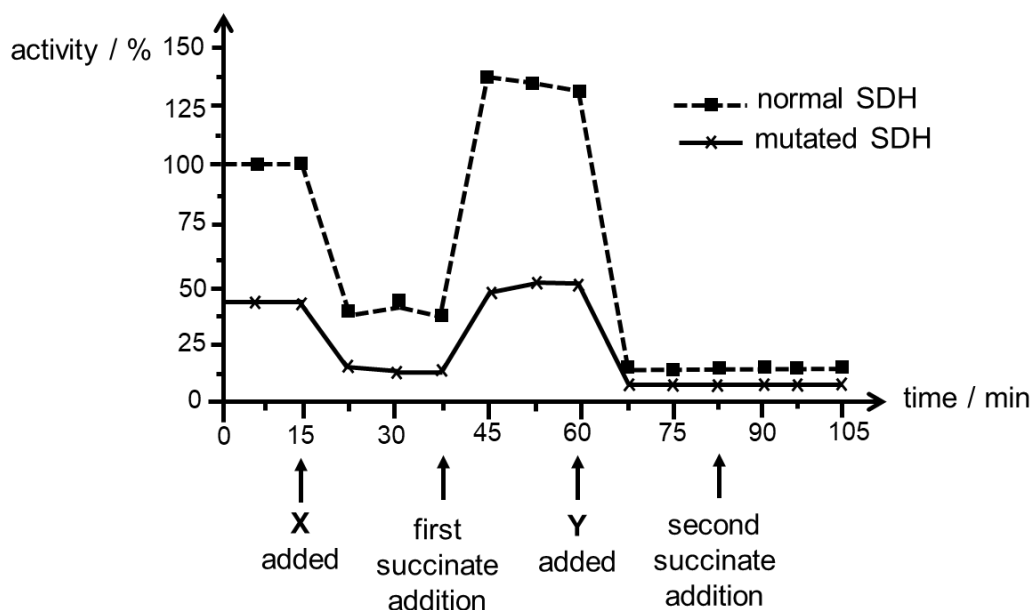
One feature of cancer cells is dysfunctional mitochondria, which is often due to mutations found in the genes coding for succinate dehydrogenase (SDH). SDH is a heterotetrameric protein complex, embedded on the inner mitochondrial membrane.

SDH is an important protein for both the Krebs cycle and oxidative phosphorylation. It catalyses the conversion of succinate to fumarate, in a reaction that is coupled with the reduction of FAD to FADH₂. SDH is also a member of the electron transport chain.

- (d) Describe how one of the subunits of the SDH complex is synthesised from its mRNA.

[5]

Inhibitors **X** and **Y** are known to affect SDH's role in the Krebs cycle. Fig. 1.2 shows changes to the activities of normal SDH and mutated SDH following the addition of inhibitor **X**, inhibitor **Y**, and an excess of succinate at different time points.



Modified from <https://cancerandmetabolism.biomedcentral.com/articles/10.1186/2049-3002-2-21>

Fig. 1.2

(e) With reference to Fig. 1.2,

- (i) account for the changes in activity of normal SDH following the addition of inhibitor **Y** and the second addition of succinate, [4]

- (ii) deduce **one** similarity in the structure of mutated and normal SDH. Justify your answer. [3]

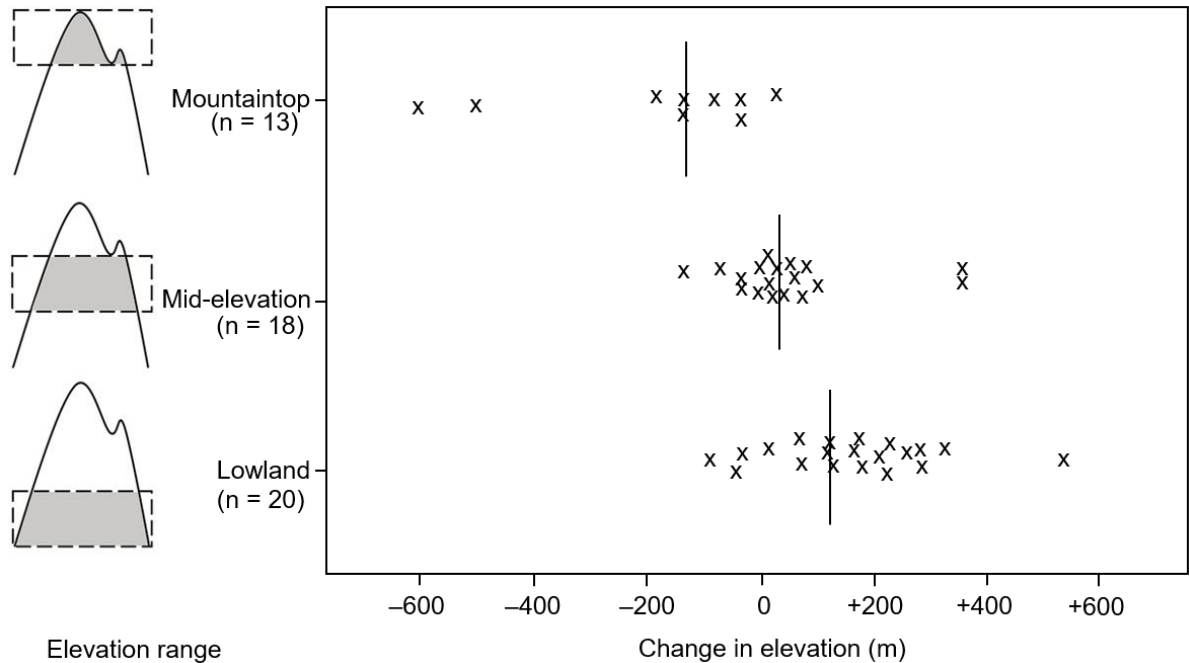
SDH is encoded by genes found in the nuclear DNA while some of the other electron carriers in the mitochondria are encoded by genes found in the mitochondrial DNA (mtDNA).

- (f) Apart from size, describe how nuclear DNA is structurally different from mtDNA. [3]

[Total: 25]

- 2 Researchers followed 51 species in Malaysia over a 10-year period to assess the impact of increasing temperature on their habitat range on the mountain. Different species respond to increasing temperature differently, including migrating to a different elevation on the mountain.

Fig. 2.1 shows the change in elevation of each species' habitat at the different elevation range on the mountain after the 10-year study.



key

n = number of species originally present at that elevation range

x = data point for change in elevation for one species after the 10-year study

I = the average change at that elevation range

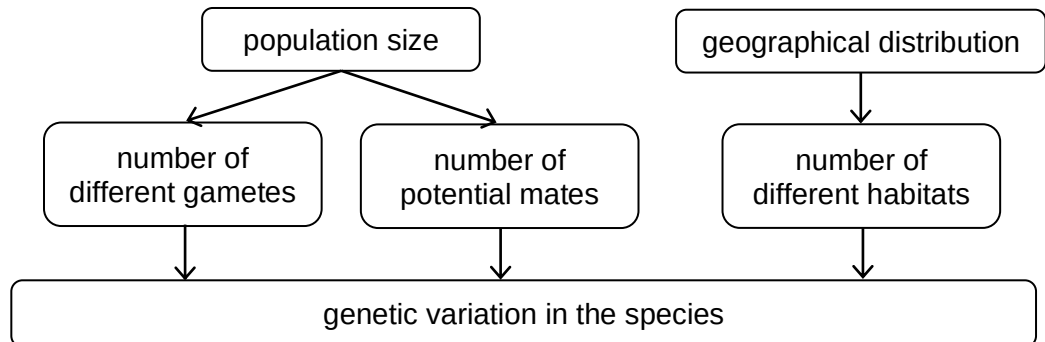
Modified from Belshaw et al. Journal of Virology. Vol.84 2010

Fig. 2.1

- (a) With reference to Fig. 2.1, describe **two** differences in how species in different elevation ranges have responded to the increasing temperature. [2]

Genetic variation in a species is key to its survival in changing environments.

Fig. 2.2 shows the relationship between population size, geographical distribution and the amount of variation present in a species.



Modified from Loughheed et al. Molecular Ecology.17(5) March 2008

Fig. 2.2

- (b) With reference to Fig. 2.2, explain how genetic variation is affected by an increase in population size and geographical distribution of a species. [4]

Unlike animals, plants have a more limited migration range. Therefore plants often have to adapt to changes in their habitat's environment.

The gene coding for aquaporin is highly varied in plant species.

Molecular and phylogenic analyses have traced the ancestral aquaporin gene to an early plant species. However, studies identified copies of the aquaporin gene present on different chromosomes in the genome of a descendant plant species.

- (c)** Suggest how mutations may have resulted in the distribution of aquaporin genes in the genome of a descendant plant species. [2]

Aquaporins are integral membrane channel proteins. They play an important role in regulating the selectivity and permeability of water molecule movement across membranes.

Fig. 2.3 shows the structure of aquaporin. Aquaporin contains five loops that join transmembrane domains I to VI. In the primary sequence of aquaporin, the amino acid sequence asparagine-proline-alanine, also known as NPA, is highly conserved in the B and E loop.

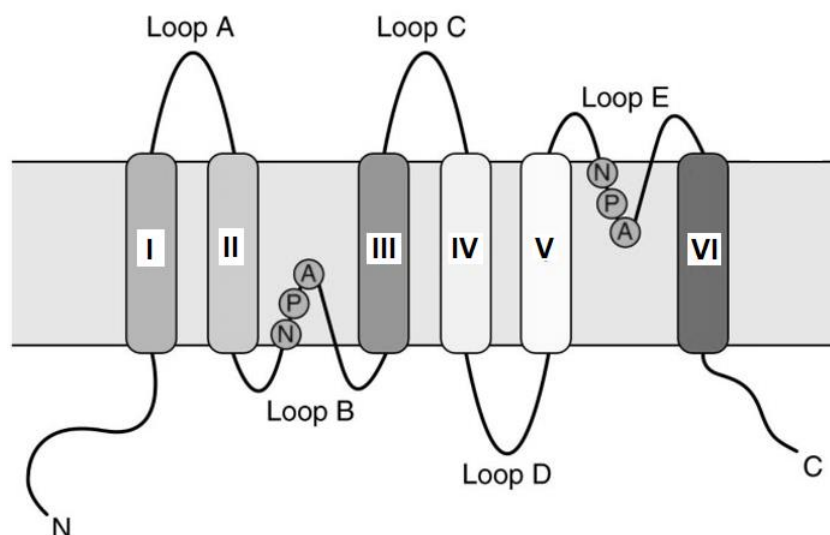
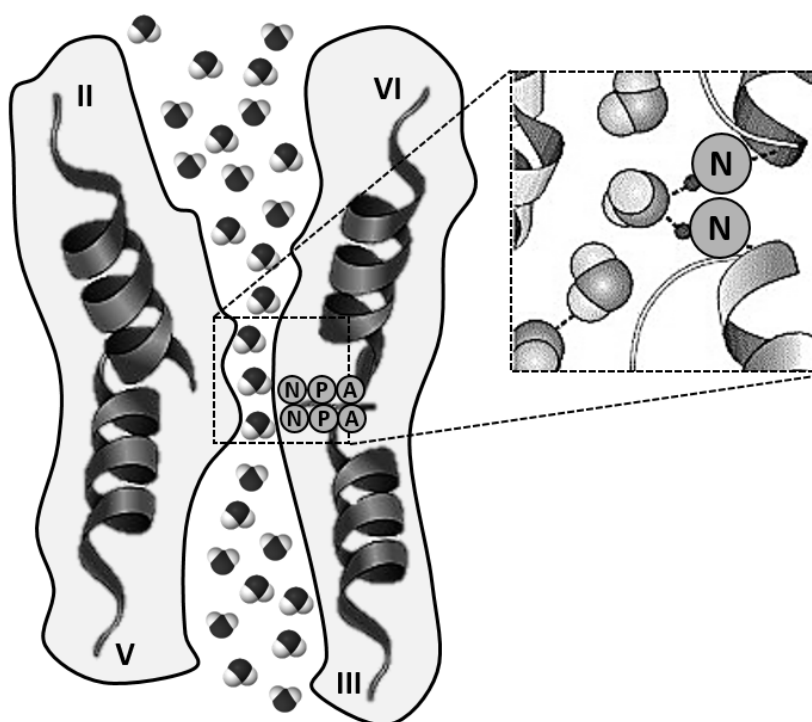


Fig. 2.3

The NPA sequences are present in the narrow central constriction of the channel as shown in Fig. 2.4.



Modified from: Kruse et al. *The aquaporins*. *Genome Biol* 7, 206 (2006)

Fig. 2.4

Fig. 2.5 shows the structure of NPA.

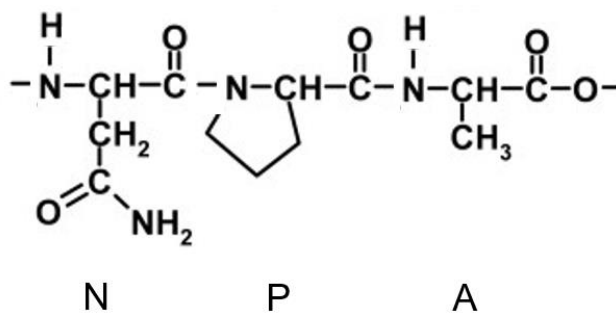


Fig. 2.5

- (d) (i) Outline how the NPA sequences come together in the aquaporin channel. [2]

- (ii) Explain how NPA regulates the selectivity of aquaporin for water molecules. [3]

- (e) At higher temperatures, both membrane fluidity and permeability increase. [2]

Explain how plants adapt to higher temperatures to reduce membrane fluidity and permeability.

[Total: 15]

- 3** Embryonic stem cells (ESCs) are a potential source of medical therapy. To overcome ethical issues related to the use of ESCs, embryonic-like pluripotent stem cells can be generated

from somatic cells by using either somatic cell nuclear transfer (SCNT) or by inducing formation of pluripotent stem cells (iPSCs). Fig. 3.1 outlines the two approaches.

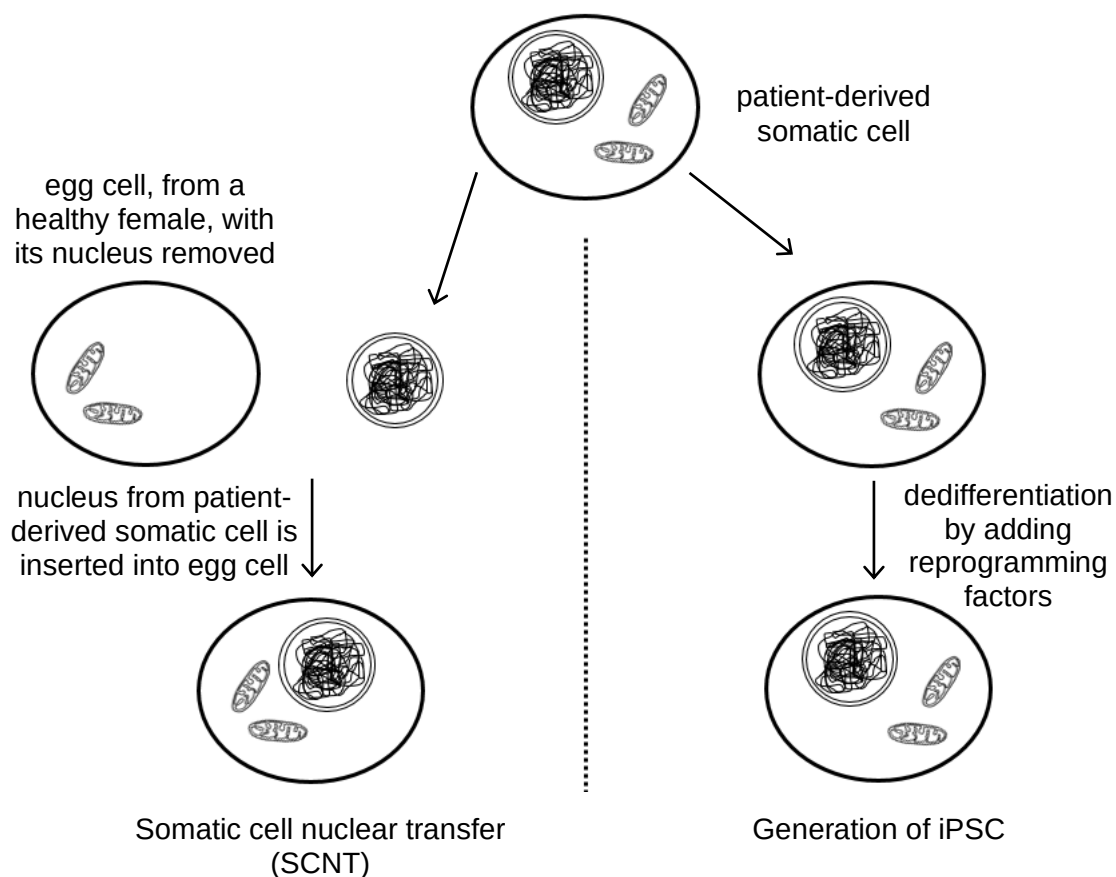


Fig 3.1

- (a) (i) Describe the normal role of embryonic stem cells. [2]

- (ii) Using your knowledge and the information provided, fill in the table below by suggesting how the use of iPSCs for therapy has an advantage or disadvantage compared with the other stem cell types. [2]

Table 3.1

	iPSCs	
compared with	advantage	disadvantage
ESCs		
SCNT-derived stem cells		

In normal development, maternal mRNAs play an important role in the early development of an embryo.

- Maternal mRNAs are synthesised by transcription but are translationally repressed during the formation of oocytes. After entering the cytoplasm, maternal mRNAs immediately undergo deadenylation, are bound by regulatory proteins at the 5' UTR region and organised into membrane-less structures known as cytoplasmic granules.
- After fertilisation between the oocyte and sperm, a period of transcriptional silence occurs. During this period, development is driven by maternal mRNAs deposited in the zygote by the oocyte.
- Zygotic genome becomes transcriptionally active as maternal mRNAs are degraded. This is known as maternal to zygotic transition (MZT).

(b) (i) Explain how maternal mRNAs are translationally repressed. [2]

(ii) Suggest how the organisation of maternal mRNAs in cytoplasmic granules allow them to be stored. [1]

During MZT, the activity of maternal mRNA leads to a positive feedback, in the activity of the zygotic mRNA, that results in its degradation.

Fig. 3.2 shows the changes in transcript levels of maternal mRNA and zygotic mRNA during MZT.

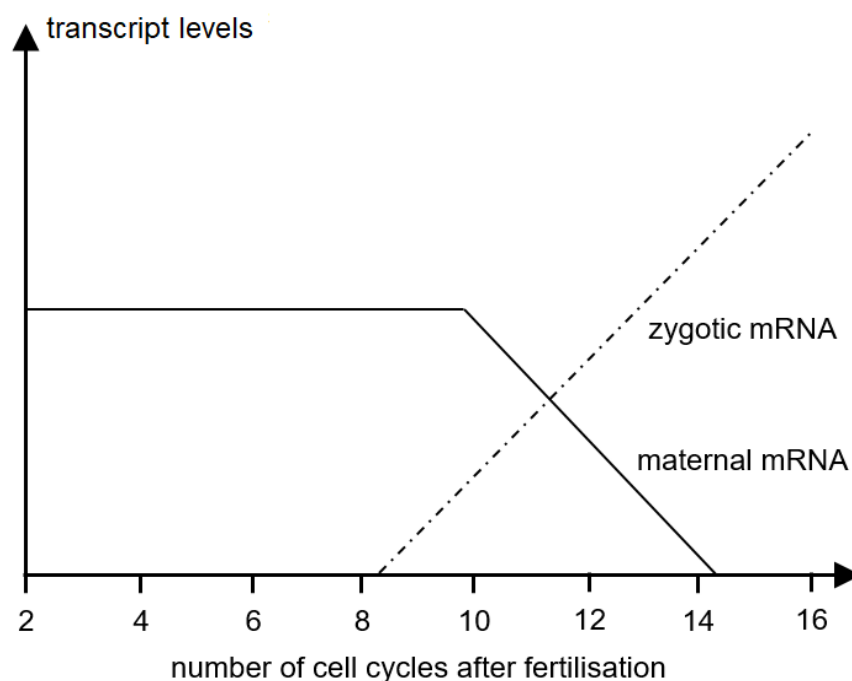


Fig. 3.2

Table 3.2 shows the changes in expression of selected enzymes that result in the change in transcript levels during MZT. Of these enzymes, only one is encoded by the maternal mRNA.

Table 3.2

	number of cell cycles from fertilisation							
enzymes	2	4	6	8	10	12	14	16
decapping enzymes				++	++	++	+	
DNA demethylation enzymes			+++	++	++	+	+	
nuclease					++	++	++	+

key

+ low rate of expression

++ moderate rate of expression

+++ high rate of expression

- (iii) With reference to Fig. 3.2 and Table 3.2, explain how expression of maternal mRNAs results in its degradation.

[3]

[Total: 10]

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 4 (a) Outline how glucagon in the bloodstream returns blood glucose concentration back to set point and describe the advantages of cell signaling pathways in multicellular organisms. [15]
- (b) Explain how features of proteins make them suited for their roles in cell signaling pathways. [10]

[Total: 25]

- 5** **(a)** Using a named example, describe how a bacteria can cause human diseases and discuss how penicillin can eliminate gram-negative bacteria. [15]
- (b)** Explain how features of biological membranes facilitate immune response in humans. [10]

[Total: 25]

[illegible]

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